

KHURAI'S PRODUCTION ENGINEERING DIVISION

South Admin Building # 2099, Room E-113

UDHAILYAH

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March 03, 2015

KhPED-116-2015

WELL SECUREMENT PROGRAM USING TTBP **ABJF-610 (PWI, HORIZONTAL)**

SAP Notification

10981988

SUPERINTENDENT

Khurais & CA Well Services Division

SUPERINTENDENT

Khurais Producing Operations Division

NOTES:

- This program also serves as a job request to SACA to issue a service order.
- This well is located 350 Km from Dhahran main gate.
- This is part of 2015 PE invoice job requirements.

OBJECTIVE:

Secure ABJF-610 by setting Through Tubing Bridge Plug (TTBP) in the 7" liner and capping it with 8 ft of cement plug.

JUSTIFICATION:

Flowmeter survey on 7/31/2010 indicated downhole leak at the depth of 3,160' with a cross flow of 3.3 MBD from ARBD to leak point (Buwaib). Arab-D formation. Setting of downhole plug is required to secure the well till the workover rig is deployed at the well to fix the 7" casing.

Note: - 1. The temperature profile now shows that there is no or minimum cross flow (see attachment)
2. 7" EZSV plug cannot be deployed as 4.5" is the maximum drift size which could pass upto 5170 ft.

History:

11/17/2007: Drilled and completed as Horizontal PWI.

01/12/2009: Acid stimulation was successfully completed on this well using 2-3/8" coiled tubing by pumping 26 stages acid systems containing a total of 2,910 bbls of 20% HCl.

07/31/2010: Downhole casing leak. Flowmeter survey by Abq Wireline indicated an anomaly at 3160 ft. with approximate cross flow rate of 3.3 MBD from reservoir to leak point.

09/14/2010: USIT and METT logs completed. Logs confirmed leak at 3170 ft.

12/24/2014: Prior to setting EZSV plug, 6.15 and 6" GC could not pass below the leak depth.

02/24/2015: Aramco Wireline run different size of GC and only 4.5" GC could pass to 5170'. 5.5" and 5" stopped at 3152.

The well is currently shut-in and waiting for workover.

CONTACT PERSONS:

KHRS & CA Well Services Section FAC	03-578-4403/0558220906
KHRS & CA Well Services Unit	578-4133 (O), 0558220908 (M)
KHRS Field Services Unit Foreman	578-4642 (O), 0550505816 (M)
KHRS SCADA	578-6138 (O)
Assistant Superintendent KhPOD	578-4177 (O)/ 505875122 (M)
CA&KHRS WIU Engineers, Mohammad A. Eissa	578-4180 (O), 0500147353 (M)
CA&KHRS WIU Senior Engineer, Sajid Mehmood	577-67885(O), 0530635689 (M)
CA&KHRS WIU Supervisor, Salman D. Gamber	577-6832 (O), 0504857948 (M)
SAPED Well Intervention Specialist, Rondall L. Zbitowsky	576-6306 (O), 0506875351 (M)
110 Emergency	03-572-0110
Khurais Ambulance/Clinic	03-578-0600

SAFETY:

• Kill fluid volume: 33,000 gals	• Kill fluid density : 75 PCF (Based last SBHP survey). Kill fluid density will be calculated based on latest SIWHP before planned well killing.
• Fluid in hole: Seawater	• H2S content: +/- 50 ppm
• BHT: 118 °F @ 5,200 ft	• Maximum dog leg: 6.63 4Deg @ 5,705' MD
• Maximum deviation: 91.3 @ 8,810' (CV)	
• Refer to Saudi Aramco Onshore Wireline Manual	
• Refer to OIM 1.148 Isolation Barriers for Wells During Producing Operations	
• Refer to OIM 1.045 Isolation Barriers for Wells During Rig-less Wireline Operations	

EQUIPMENT REQUIREMENTS**Service Company will provide**

- Complete E-line logging setup including logging truck and pressure control equipment.
- Through Tubing Bridge Plug (TTBP) suitable for 1500 psig differential pressure with setting tool. A back-up TTBP should be available on location.
- Dump bailer and cement kit and cement testing equipment.
- PCE as per Saudi Aramco Wireline Manual
- Required crane and related equipment.
- Recently calibrated pressure recorders.
- Necessary wellhead connections and flanges.
- Standard safety equipment as specified in the safety guide lines for e-line jobs.
- One suitable pumping unit for performing pressure tests.

KHRS&CA WSD Well Services Unit:

- Standard safety equipment as specified in Slick Line Safety Checklist.
- One 10,000 gal of Jilh water tanker on standby for pressure testing and pumping.
- Three sets of floodlights to be used if needed.

Please note that these requirements were distributed based upon OIM # 1.047 (ROLES & RESPONSIBILITIES OF OIL & WATER WELLS IN S. A. PRODUCING). For more details, please refer to the subject OIM.

PRE-JOB REQUIREMENTS**KHRS&CA Field Activity Coordinator**

- Ensure all pre-job requirements are completed before the job date.
- Arrange a pre-job meeting with Service Company, Well Services and KhPED one week prior the job date.
- Conduct Hazard Assessment meeting with KhPED and Service Company prior to executing the job.
- Coordinate with KhPOD/FSU Foreman to prepare the well for the job.
- Consult KhPED prior to mobilizing Service Company to the site.

NG/KHRS&CA WSD Well Services Unit

- Ensure the location is accessible for the equipment mobilization.
- Ensure that all equipment and material are available on location one day before the job.
- Well Services Foreman will be responsible for all operation/decision making on the well site including safety and well control.

- Prepare the well for rigging up the wireline/ slickline lubricator to run the Gauge Cutter and plugs, including replacing the Crown Valve (CV), if required.
- Replace the existing 3" CV with 7-1/16" 3M Gate Valve (API spec) on X-Mas tree, if required.
- Ensure that program is followed "as is" to obtain technical objectives. Any significant deviation from program has to be discussed with KhPED and a supplement to the program shall be issued by KhPED.

Khurais Field Services Unit

- Conduct Wellhead integrity test at minimum 2800 psig and pass on the results to PE and FAC immediately. The plug setting job will be conducted only if the Master Valve is found to be in good condition. Also conduct annuli survey one week before the job.
- Ensure the spectacle blind on kill line is in open position to bleed the pressure to pit, if needed.

KHRS Wireline

- RIH 4.5" gauge cutter (GC) to **approximately 5,100'**. If held-up is encountered or GC fails to reach the target depth, stop and discuss the findings and running options with KhPED Engineer.

KhPE Engineer

- Provide the well data to KHRS&CA WSU Foreman, if requested.
- Witness the operation at site and provide technical support to KHRS&CA WCU Foreman, if requested.

Service Company

- Service Company must carry out a lab test for the cement to be used for the job. The lab result will be checked by PE witness prior to job start.
- Produce relevant certificate showing the lab test done for the cement to be used and visit the well prior to the scheduled date to ensure that there are no impediments to rig up on the well.
- Visit the well site one week before the scheduled job date to ensure that there are no impediments to rigging up on the well.
- Obtain the correlation logs from RDD library (USIT/GR 9/14/2010).
- Install temporary reflective signs on the highway and access roads to show the way to well location.
- Prepare a dimensional drawing of wellhead, riser, BOP's, crossovers and labeled and dimensioned bottom-hole assembly (BHA) drawing for discussion during the pre-job meeting.
- Prepare the usual procedure for setting, retrieving the plugs, including contingency plan.
- Provide the job data event in electronic readable format (excel format). Prepare a job completion report with results and submit to KhPED WIU within 3 days after the job completion.

PROCEDURE:

General Data Sheet

ABJF-610

9 5/8" Casing

: Weight 40 lb/ft, Grade J-55, ID 8.835"

7" Casing

: Weight 26 lb/ft, Grade J-55, ID 6.276"

7" liner top at 732 ft.

4.5" GC could run up to 5170 ft.

Pre-checks:

- **Check cement lab test analysis result from Service Company**
- **Check the cement specifications including expiry date and additives**
- **Request Service Company to explain what additives to be used for the downhole condition.**

1. Perform a pre-job safety audit as per the attached wire line Safety Checklist and record SIWHP and CCA Pressure.

2. Hold pre-job safety meeting to discuss procedures and potential hazards of the job.
3. Rig up BOP on top of crown valve. Function and pressure test BOP and other PCE to 2500 psi as per attachment. (Refer to Saudi Aramco Wireline manual for details).
4. Make up and function test BHA.
5. Rig up riser on top of BOP (Ensure BOP and riser pipe minimum restriction is greater than BHA maximum outside diameter). Pressure test riser to 2500 psi against wellhead valves. Calibrate & function test tools and perform surface checks. (pressure test after every break-up and make up)
6. With the well shut-in, RIH BHA with a TTBP capable of passing through the minimum restriction in in the well as per GC results.
7. Confirm the depth of the TTBP by running a CCL log. Set the plug at 4500 ft.. Setting depth should be adequately away from casing collars. RIH and tag top of the plug to confirm the plug setting depth. POOH slowly with the plug setting tool.
8. Mix a HIP cement kit. Check the Quality Control specification sheet included with the cement kit and ensures that the cement kit is not beyond its shelf life. Provide cement sample for each mix to KhPED engineer. Another set of sample to be retained by Service Company for further lab testing and report result to PE within 5 days of operation.
9. RIH slowly with the 2-1/8" cement bailer and dump 8' of cement on top of each plug. POOH.
10. Rig down Service Company equipment and secure the well. Secure wellhead and report job completion to KCPF operations Foreman.
11. Service Company to submit a job detailed report including logs and cement lab result within 5 days of completion of job.

Originally signed by SDG

Salman D. Gamber KhPE Water Injection Unit

NKS

Attachments: (BOP Test Procedure, Well X-Section, WH Diagram, Deviation Survey, Well Location Map, Safety Guidelines for Slick line Jobs, Basic Pre-Intervention Well Data, Tools Check List, PCE Checklist, and Minimum Well Control Requirements, Typical Slick Line rig-up for water Wells)

CC:

Supervisor, KhPED/WIU
SAPED Well Intervention Specialist (ZBITOWRL)
KhPD KhPFOD Foremen
Foreman, KhPOD/KHRS Field Services Unit
Foreman, KHRS / CA Well Completion Unit
Foreman, KHRS Wireline Unit
KhPED/WIU Engineers; aldhafkg
FAC, KHRS / CA Well Services Section;
E-Well File:ABJF-610

ATTACHMENT-1: WIRELINE BOP TEST REQUIREMENT

Testing Wireline BOP Equipment

BOP's shall be pressure tested at the well site before every job, quarterly and after maintenance in the shop. Annual certification of the equipment SHALL be required as described in Section 6.2. (Saudi Aramco Wireline Manual)

The Saudi Aramco operations use a variety of wireline sizes so pressure and function testing of each BOP system at the well site, preferably before it is placed on the tree, SHALL be required. This provides assurance that the BOP rams are properly sized for the cable being used and that they will operate as designed when required (usually an emergency or risky remedial operation). In order to test the BOP prior to installation on the tree, ensure that a "stump" test assembly with the appropriate blank testing subs and plates that prevent rod movement are available.

The pressure testing of the ram seals shall be done with a machined and calibrated diameter rod that is equivalent to the wireline diameter as per the manufacturer's recommendation. Do NOT use wireline to perform the testing because it will invariably deform or damage the elastomer and may not provide a definitive pressure test due to leakage through the wireline armor "gaps". This testing will also provide a check on the ram functionality and the accumulator capacity and capability.

The smooth BOP test rods will be precisely machined and then used in a testing system with the following design features:

- Use a specially design "stump" test rack with appropriate design features that will minimize personnel and equipment safety risks while performing the testing procedures
- Fit a handle on the rod so it can be lifted in and out of the BOP. A cross piece is attached to prevent the rod from being dropped through the BOP
- Sealing surface is smooth and machined to match the diameter of the cable in use
- Screw a plate on the bottom of the rod and back it up with a nut screwed directly underneath. The plate should be located just under the rams during a pressure test to prevent the rod from being blown out.

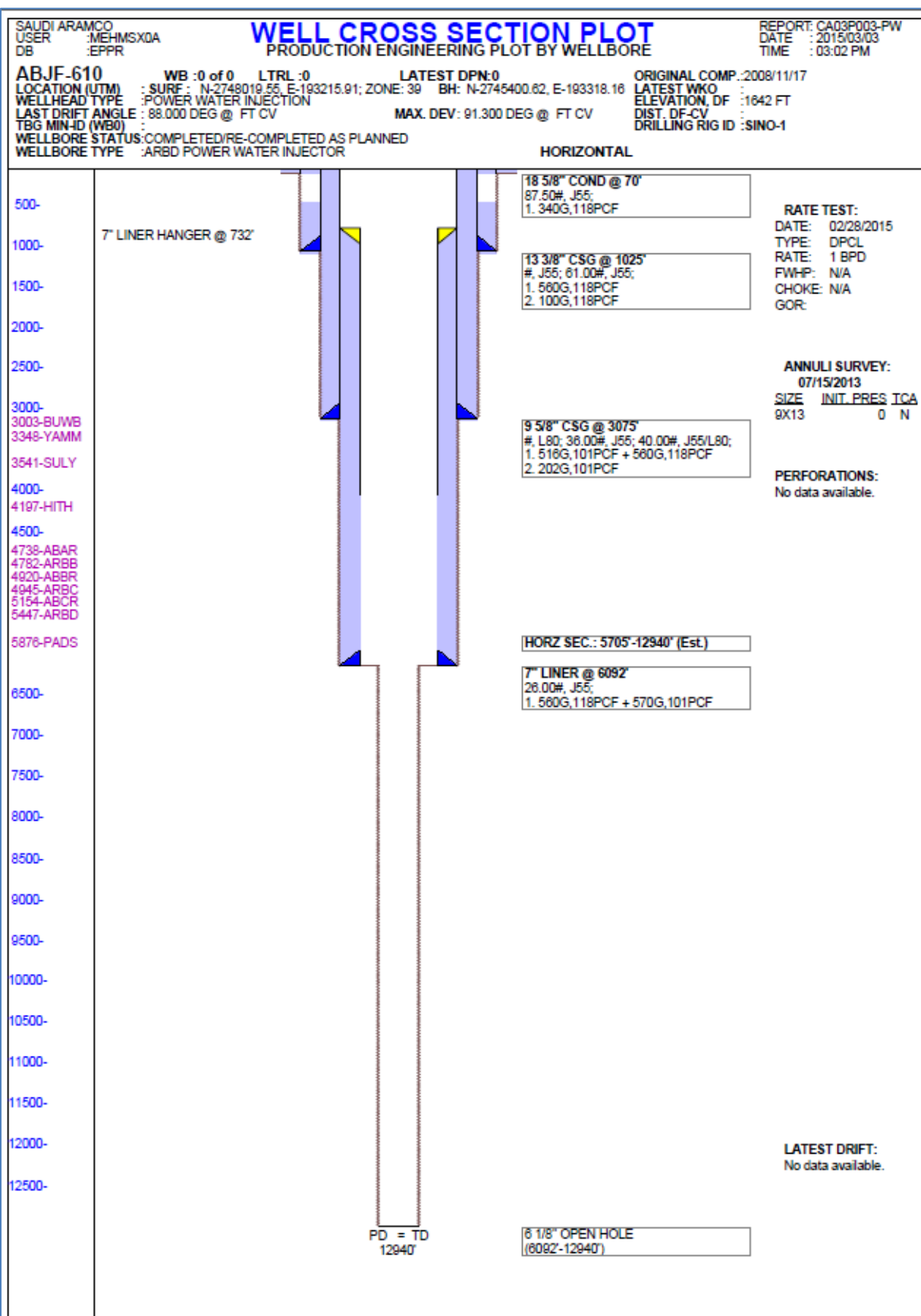
BEST PRACTICE

Pressure and function test wireline BOP rams with a machined rod (Rod Test) before it is installed on the tree. This should be done with a specially designed test rack – sometimes called "stump" testing.

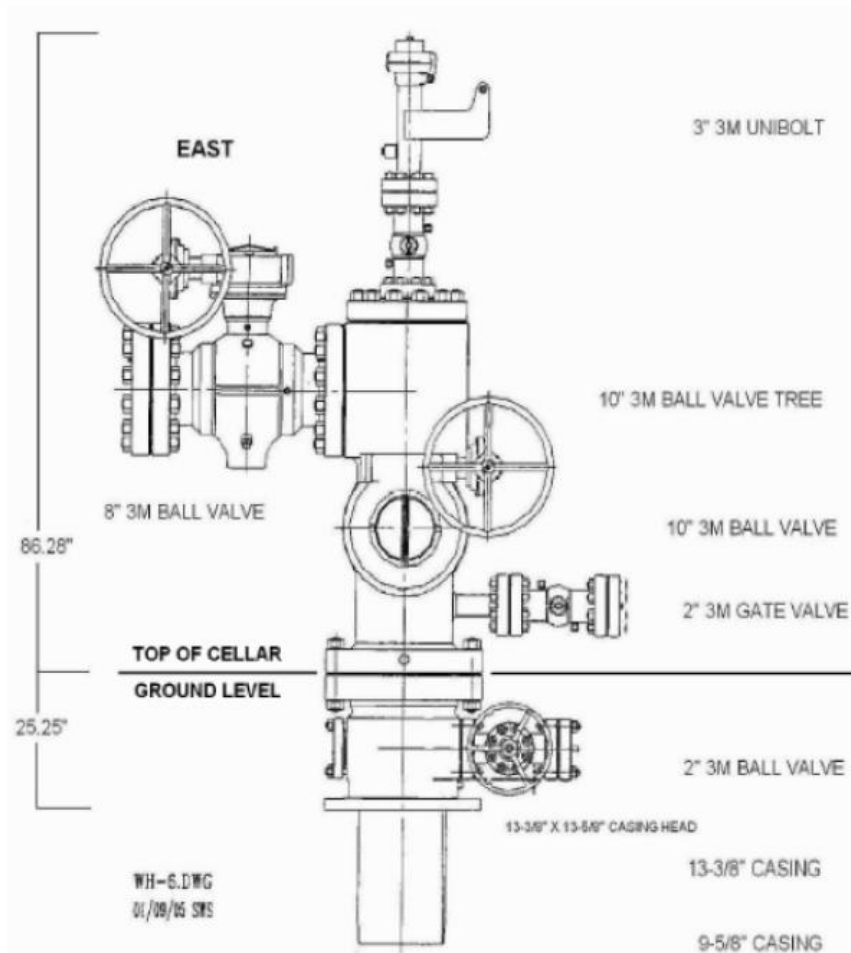
A Rod Test SHALL be performed prior to every job (different well) to the Equipment Working Pressure (WP).

A body pressure test SHALL be performed each time prior to running in the hole to the Equipment Working Pressure (WP).

Note: Refer to Saudi Aramco onshore wireline manual for details



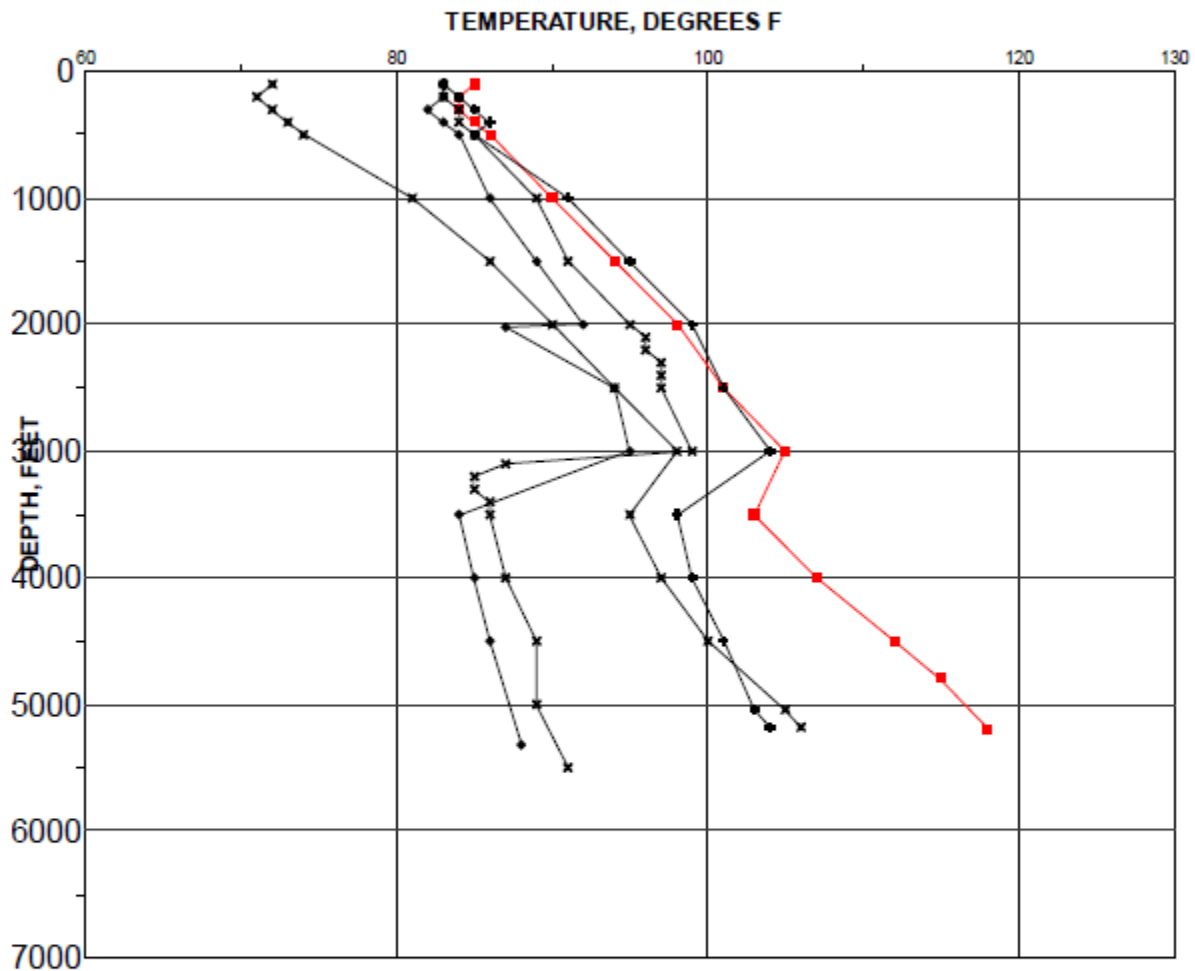
Wellhead diagram ABJF-610



Temperature Profile

Report : TMTRR002 Database : EPPR User : MEHMSX0A Run Date & Time : 3/3/2015 16:18	Temperature Survey Plot	 ارامكو السعودية Saudi Aramco Page 1 of 3
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FIELD : ABJF WELL : 610 TOTAL DEPTH : 12940	GOSP/WIP : KHR50201 (KHR5 PWI) WELL TYPE : POWER WATER INJECTOR REPORT PERIOD : 03/04/2010 - 07/23/2014	
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SYMBOL	SURVEY DATE	RN LOC	TYPE	G MOV	BMB NO	SI HRS.	CSG SI-WHP	TBG SI-WHP	VALIDITY	LTRL/(WB)
■	07/23/2014	1 BOT	SHUT-IN	DOWN	0001	999			VALID SURVEY	0/(0)
×	11/07/2012	1 BOT	SHUT-IN	DOWN	0001	999			VALID SURVEY	0/(0)
+	04/04/2011	1 BOT	SHUT-IN	DOWN	0001				VALID SURVEY	0/(0)
×	04/12/2010	1 BOT	SHUT-IN	DOWN	0001	48			VALID SURVEY	0/(0)
♦	03/04/2010	1 BOT	SHUT-IN	DOWN	0001				DISCARDED (NO	0/(0)

DEVIATION SURVEY REPORT BY FIELDWELL/LATERAL

FIELD AND WELL : ABJF-610				(WB : 0)				MAX: DRIFT ANGLE : 91.300				SURFACE LOCATION NORTHING : 2748019.55			
BOT. LATERAL NAME : 0				TIE-IN-POINT : 936269				DEPTH (DF-FT) : 8810				EASTING : 193215.91			
TIE-IN-DEPTH : 0								VALID FLAG : Yes				ZONE : 39			
SERVICE COMPANY : SPERRY SUN								AAP FLAG : Yes				REF ELEV & CODE : 1642 DF			
LATERAL NAME	SURVEY DATE	MEAS DEPTH	TVD (DF)	TVD (SS)	DRIFT ANGLE (DEG)	TOOL DIRECT. (DEG)	DOGLEG (DEG/100 FT)	Y-OFFSET (METERS)	X-OFFSET (METERS)	TOOL TYPE	SURV. VERS.	DSPT NUM			
0	2008/11/17	0	0	1642	0.000	0.000	0.00	0.000	0.000	0.000 MULTI SHOT	AAP	936269			
0	2008/11/17	81	81	1561	0.920	279.130	1.14	0.031	-0.196	MULTI SHOT	AAP	936270			
0	2008/11/17	172	172	1470	0.870	345.970	1.08	0.271	-0.467	MULTI SHOT	AAP	936271			
0	2008/11/17	262	262	1380	0.760	340.960	0.15	0.645	-0.576	MULTI SHOT	AAP	936272			
0	2008/11/17	353	353	1289	0.770	341.120	0.01	0.995	-0.697	MULTI SHOT	AAP	936273			
0	2008/11/17	444	444	1198	0.760	341.410	0.01	1.346	-0.816	MULTI SHOT	AAP	936274			
0	2008/11/17	534	534	1108	1.100	338.800	0.38	1.764	-0.969	MULTI SHOT	AAP	936275			
0	2008/11/17	625	625	1017	1.110	331.180	0.16	2.248	-1.195	MULTI SHOT	AAP	936276			
0	2008/11/17	715	715	927	1.190	324.780	0.17	2.713	-1.487	MULTI SHOT	AAP	936277			
0	2008/11/17	804	804	838	1.240	325.460	0.06	3.185	-1.816	MULTI SHOT	AAP	936278			
0	2008/11/17	881	881	761	1.260	324.630	0.03	3.605	-2.109	MULTI SHOT	AAP	936279			
0	2008/11/17	974	974	668	1.320	327.080	0.09	4.133	-2.467	MULTI SHOT	AAP	936280			
0	2008/11/17	1063	1063	579	1.020	315.190	0.43	4.566	-2.807	MULTI SHOT	AAP	936281			
0	2008/11/17	1153	1153	489	0.700	297.620	0.46	4.817	-3.128	MULTI SHOT	AAP	936282			
0	2008/11/17	1244	1244	398	0.570	289.750	0.17	4.942	-3.408	MULTI SHOT	AAP	936283			
0	2008/11/17	1335	1335	307	0.690	290.720	0.13	5.048	-3.694	MULTI SHOT	AAP	936284			
0	2008/11/17	1425	1425	217	0.940	297.350	0.30	5.210	-4.048	MULTI SHOT	AAP	936285			
0	2008/11/17	1515	1515	127	1.040	299.990	0.12	5.438	-4.464	MULTI SHOT	AAP	936286			
0	2008/11/17	1610	1610	32	1.610	302.170	0.60	5.786	-5.035	MULTI SHOT	AAP	936287			
0	2008/11/17	1705	1705	-63	2.080	319.650	0.77	6.403	-5.720	MULTI SHOT	AAP	936288			
0	2008/11/17	1800	1800	-158	2.190	331.010	0.46	7.287	-6.328	MULTI SHOT	AAP	936289			
0	2008/11/17	1895	1895	-253	2.120	333.850	0.13	8.252	-6.833	MULTI SHOT	AAP	936290			
0	2008/11/17	1990	1990	-348	1.640	342.950	0.59	9.129	-7.190	MULTI SHOT	AAP	936291			
0	2008/11/17	2084	2083	-441	1.380	345.270	0.28	9.854	-7.398	MULTI SHOT	AAP	936292			
0	2008/11/17	2179	2178	-536	1.490	5.990	0.56	10.566	-7.447	MULTI SHOT	AAP	936293			
0	2008/11/17	2273	2272	-630	1.660	23.960	0.55	11.316	-7.240	MULTI SHOT	AAP	936294			
0	2008/11/17	2367	2366	-724	2.090	40.740	0.74	12.091	-6.731	MULTI SHOT	AAP	936295			
0	2008/11/17	2462	2461	-819	2.010	54.310	0.52	12.787	-5.974	MULTI SHOT	AAP	936296			
0	2008/11/17	2557	2556	-914	1.900	63.800	0.36	13.295	-5.130	MULTI SHOT	AAP	936297			
0	2008/11/17	2651	2650	-1008	2.090	78.810	0.59	13.606	-4.192	MULTI SHOT	AAP	936298			
0	2008/11/17	2747	2746	-1104	1.860	76.840	0.25	13.818	-3.206	MULTI SHOT	AAP	936299			
0	2008/11/17	2841	2840	-1198	1.650	87.750	0.42	13.940	-2.341	MULTI SHOT	AAP	936300			
0	2008/11/17	2936	2935	-1293	1.450	94.650	0.29	13.927	-1.559	MULTI SHOT	AAP	936301			
0	2008/11/17	3031	3030	-1388	1.830	84.610	0.50	13.941	-0.734	MULTI SHOT	AAP	936302			
0	2008/11/17	3072	3071	-1429	2.010	81.830	0.49	13.990	-0.318	MULTI SHOT	AAP	936303			
0	2008/11/17	3149	3148	-1506	2.830	158.210	3.97	13.511	0.304	MULTI SHOT	AAP	936304			
0	2008/11/17	3245	3244	-1602	6.550	168.140	3.95	11.206	0.916	MULTI SHOT	AAP	936305			
0	2008/11/17	3340	3338	-1696	8.210	178.400	2.22	7.523	1.313	MULTI SHOT	AAP	936306			

DEVIATION SURVEY REPORT BY FIELDWELL/LATERAL

FIELD AND WELL : ABJF-610				(WB : 0)				MAX: DRIFT ANGLE : 91.300				SURFACE LOCATION NORTHING : 2748019.55			
BOT. LATERAL NAME : 0				TIE-IN-POINT : 936269				DEPTH(DF-FT) : 8810				EASTING : 193215.91			
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SERVICE COMPANY : SPERRY SUN								AAP FLAG : Yes				REF ELEV & CODE : 1642 DF			
LATERAL NAME	SURVEY DATE	MEAS DEPTH	TVD (DF)	TVD (SS)	DRIFT ANGLE(DEG)	TOOL DIRECT. (DEG)	DOGLEG (DEG/100 FT)	Y-OFFSET (METERS)	X-OFFSET (METERS)	TOOL TYPE	SURV. VERS.	DSPT NUM			
0	2008/11/17	3435	3432	-1790	8.600	178.550	0.41	3.292	1.426	MULTI SHOT	AAP	936307			
0	2008/11/17	3530	3525	-1883	11.510	177.760	3.07	-1.760	1.593	MULTI SHOT	AAP	936308			
0	2008/11/17	3624	3617	-1975	14.450	174.380	3.23	-8.176	2.055	MULTI SHOT	AAP	936309			
0	2008/11/17	3719	3708	-2066	18.680	176.820	4.51	-16.405	2.666	MULTI SHOT	AAP	936310			
0	2008/11/17	3813	3796	-2154	21.830	176.670	3.35	-26.307	3.231	MULTI SHOT	AAP	936311			
0	2008/11/17	3909	3885	-2243	23.850	176.530	2.10	-37.644	3.905	MULTI SHOT	AAP	936312			
0	2008/11/17	4003	3970	-2328	25.690	177.230	1.98	-49.630	4.555	MULTI SHOT	AAP	936313			
0	2008/11/17	4097	4054	-2412	28.330	177.370	2.81	-62.626	5.168	MULTI SHOT	AAP	936314			
0	2008/11/17	4192	4137	-2495	29.370	177.840	1.12	-76.585	5.751	MULTI SHOT	AAP	936315			
0	2008/11/17	4287	4219	-2577	31.960	178.150	2.73	-91.343	6.266	MULTI SHOT	AAP	936316			
0	2008/11/17	4382	4298	-2656	35.070	176.950	3.35	-107.314	6.956	MULTI SHOT	AAP	936317			
0	2008/11/17	4475	4374	-2732	34.590	176.820	0.52	-123.480	7.836	MULTI SHOT	AAP	936318			
0	2008/11/17	4570	4451	-2809	36.870	177.630	2.45	-140.368	8.651	MULTI SHOT	AAP	936319			
0	2008/11/17	4665	4526	-2884	39.350	177.440	2.61	-158.221	9.420	MULTI SHOT	AAP	936320			
0	2008/11/17	4759	4597	-2955	42.430	177.930	3.29	-176.958	10.175	MULTI SHOT	AAP	936321			
0	2008/11/17	4854	4666	-3024	44.790	178.050	2.49	-196.917	10.875	MULTI SHOT	AAP	936322			
0	2008/11/17	4948	4731	-3089	47.840	179.060	3.34	-217.627	11.393	MULTI SHOT	AAP	936323			
0	2008/11/17	5043	4793	-3151	50.300	178.430	2.64	-239.496	11.874	MULTI SHOT	AAP	936324			
0	2008/11/17	5137	4852	-3210	52.290	178.430	2.12	-261.845	12.487	MULTI SHOT	AAP	936325			
0	2008/11/17	5232	4907	-3265	56.990	178.550	4.95	-285.445	13.108	MULTI SHOT	AAP	936326			
0	2008/11/17	5326	4956	-3314	60.370	177.240	3.79	-309.900	14.012	MULTI SHOT	AAP	936327			
0	2008/11/17	5421	4999	-3357	64.910	175.390	5.08	-335.555	15.673	MULTI SHOT	AAP	936328			
0	2008/11/17	5516	5036	-3394	69.310	176.930	4.87	-362.164	17.453	MULTI SHOT	AAP	936329			
0	2008/11/17	5564	5052	-3410	71.640	178.400	5.65	-375.940	18.014	MULTI SHOT	AAP	936330			
0	2008/11/17	5611	5067	-3425	72.110	178.430	1.00	-389.549	18.390	MULTI SHOT	AAP	936331			
0	2008/11/17	5658	5081	-3439	72.910	178.850	1.90	-403.209	18.714	MULTI SHOT	AAP	936332			
0	2008/11/17	5705	5094	-3452	75.980	178.300	6.63	-417.004	19.058	MULTI SHOT	AAP	936333			
0	2008/11/17	5752	5104	-3462	78.190	177.800	4.82	-430.958	19.533	MULTI SHOT	AAP	936334			
0	2008/11/17	5800	5113	-3471	79.800	177.360	3.47	-445.306	20.140	MULTI SHOT	AAP	936335			
0	2008/11/17	5848	5121	-3479	82.730	177.510	6.11	-459.750	20.787	MULTI SHOT	AAP	936336			
0	2008/11/17	5895	5126	-3484	83.750	177.650	2.19	-473.964	21.388	MULTI SHOT	AAP	936337			
0	2008/11/17	5942	5131	-3489	85.010	177.600	2.68	-488.208	21.978	MULTI SHOT	AAP	936338			
0	2008/11/17	5989	5135	-3493	84.100	177.700	1.95	-502.457	22.563	MULTI SHOT	AAP	936339			
0	2008/11/17	6036	5140	-3498	83.390	177.710	1.51	-516.685	23.133	MULTI SHOT	AAP	936340			
0	2008/11/17	6121	5149	-3507	84.990	177.880	1.89	-542.440	24.125	MULTI SHOT	AAP	936341			
0	2008/11/17	6167	5152	-3510	86.230	177.760	2.71	-556.410	24.657	MULTI SHOT	AAP	936342			
0	2008/11/17	6213	5155	-3513	86.290	177.980	0.49	-570.391	25.177	MULTI SHOT	AAP	936343			
0	2008/11/17	6262	5158	-3516	86.910	178.030	1.27	-585.291	25.696	MULTI SHOT	AAP	936344			

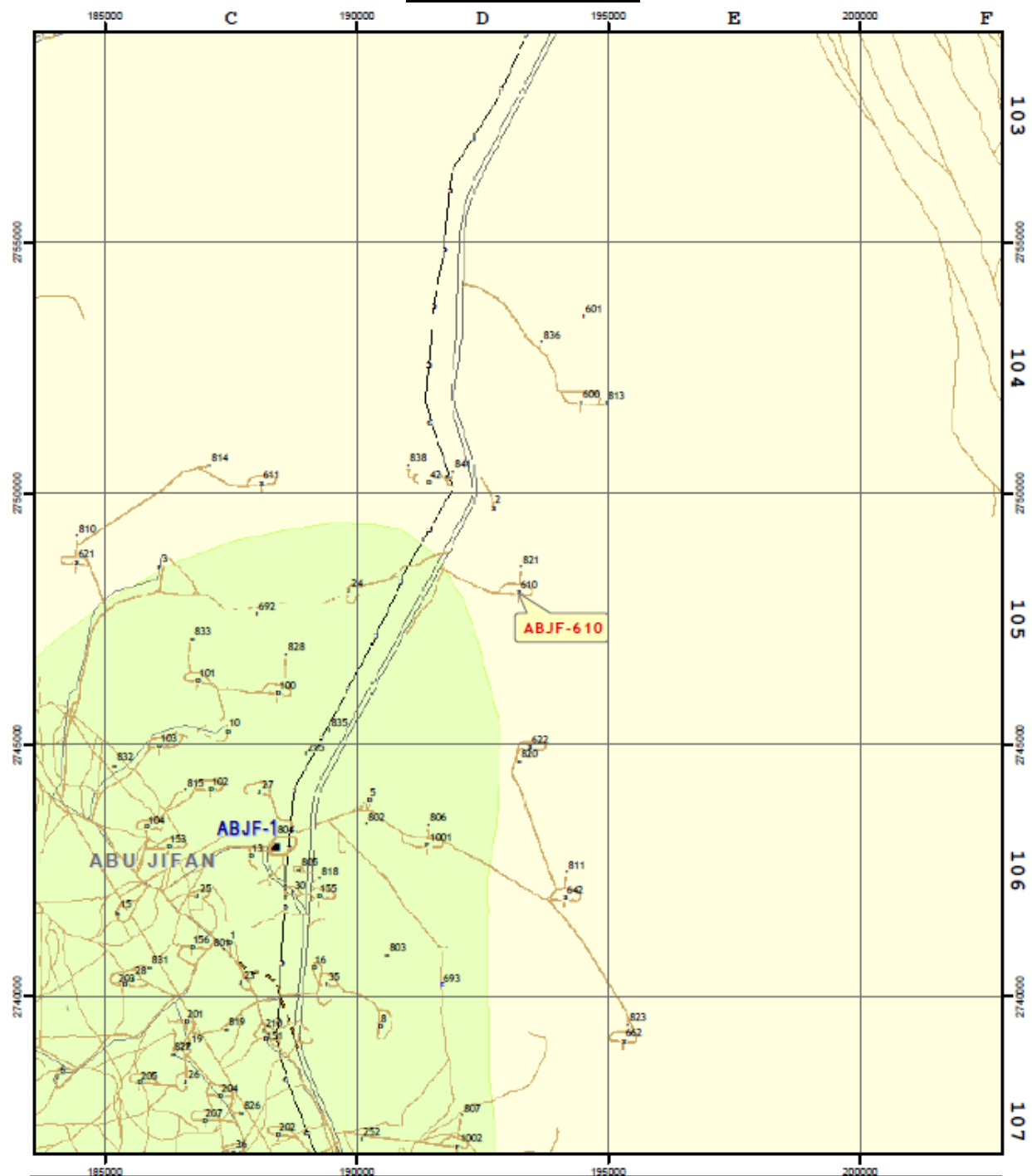
DEVIATION SURVEY REPORT BY FIELDWELL/LATERAL

FIELD AND WELL : ABJF-610				(WB : 0)				TIE-IN-POINT : 936269				SURFACE LOCATION NORTHING : 2748019.55			
BOT. LATERAL NAME : 0								DEPTH (DF-FT) : 8810				EASTING : 193215.91			
TIE-IN-DEPTH : 0								VALID FLAG : Yes				ZONE : 39			
SERVICE COMPANY : SPERRY SUN								AAP FLAG : Yes				REF ELEV & CODE : 1642 DF			
LATERAL NAME	SURVEY DATE	MEAS DEPTH	TVD (DF)	TVD (SS)	DRIFT ANGLE (DEG)	TOOL DIRECT. (DEG)	DOGLEG (DEG/100 FT)	Y-OFFSET (METERS)	X-OFFSET (METERS)	TOOL TYPE	SURV. VERS.	DSPT NUM			
0	2008/11/17	6305	5160	-3518	87.710	178.190	1.90	-598.375	26.127	MULTI SHOT	AAP	936345			
0	2008/11/17	6351	5162	-3520	89.200	178.100	3.25	-612.383	26.581	MULTI SHOT	AAP	936346			
0	2008/11/17	6397	5162	-3520	90.000	178.070	1.74	-626.396	27.050	MULTI SHOT	AAP	936347			
0	2008/11/17	6490	5163	-3521	88.760	177.820	1.36	-654.722	28.066	MULTI SHOT	AAP	936348			
0	2008/11/17	6582	5165	-3523	88.700	177.480	0.38	-682.733	29.216	MULTI SHOT	AAP	936349			
0	2008/11/17	6674	5165	-3523	90.870	177.810	2.39	-710.749	30.368	MULTI SHOT	AAP	936350			
0	2008/11/17	6765	5165	-3523	89.940	177.680	1.03	-738.463	31.459	MULTI SHOT	AAP	936351			
0	2008/11/17	6857	5164	-3522	90.560	177.960	0.74	-766.484	32.526	MULTI SHOT	AAP	936352			
0	2008/11/17	6949	5165	-3523	88.950	178.100	1.76	-794.508	33.490	MULTI SHOT	AAP	936353			
0	2008/11/17	7042	5166	-3524	89.440	178.340	0.59	-822.838	34.370	MULTI SHOT	AAP	936354			
0	2008/11/17	7134	5167	-3525	89.810	178.470	0.43	-850.868	35.151	MULTI SHOT	AAP	936355			
0	2008/11/17	7226	5167	-3525	89.750	178.420	0.08	-878.899	35.912	MULTI SHOT	AAP	936356			
0	2008/11/17	7318	5167	-3525	90.000	178.200	0.36	-906.928	36.738	MULTI SHOT	AAP	936357			
0	2008/11/17	7410	5168	-3526	88.770	177.860	1.39	-934.951	37.702	MULTI SHOT	AAP	936358			
0	2008/11/17	7502	5170	-3528	89.380	178.040	0.69	-962.971	38.705	MULTI SHOT	AAP	936359			
0	2008/11/17	7594	5170	-3528	89.750	177.900	0.43	-990.994	39.699	MULTI SHOT	AAP	936360			
0	2008/11/17	7686	5171	-3529	89.690	177.600	0.33	-1019.014	40.800	MULTI SHOT	AAP	936361			
0	2008/11/17	7778	5172	-3530	88.640	177.440	1.15	-1047.026	42.013	MULTI SHOT	AAP	936362			
0	2008/11/17	7871	5173	-3531	89.630	177.270	1.08	-1075.338	43.321	MULTI SHOT	AAP	936363			
0	2008/11/17	7963	5174	-3532	89.690	178.200	1.01	-1103.357	44.429	MULTI SHOT	AAP	936364			
0	2008/11/17	8057	5174	-3532	90.000	179.030	0.94	-1132.000	45.121	MULTI SHOT	AAP	936365			
0	2008/11/17	8151	5174	-3532	90.740	179.230	0.82	-1160.647	45.556	MULTI SHOT	AAP	936366			
0	2008/11/17	8245	5175	-3533	87.900	178.420	3.14	-1189.287	46.144	MULTI SHOT	AAP	936367			
0	2008/11/17	8340	5176	-3534	90.190	178.050	2.44	-1218.223	47.035	MULTI SHOT	AAP	936368			
0	2008/11/17	8433	5178	-3536	87.650	177.010	2.95	-1246.535	48.257	MULTI SHOT	AAP	936369			
0	2008/11/17	8527	5180	-3538	89.750	176.230	2.38	-1275.127	49.945	MULTI SHOT	AAP	936370			
0	2008/11/17	8622	5180	-3538	90.250	176.530	0.61	-1304.026	51.774	MULTI SHOT	AAP	936371			
0	2008/11/17	8716	5182	-3540	88.020	176.120	2.41	-1332.613	53.610	MULTI SHOT	AAP	936372			
0	2008/11/17	8810	5182	-3540	91.300	176.570	3.52	-1361.201	55.436	MULTI SHOT	AAP	936373			
0	2008/11/17	8904	5182	-3540	89.260	177.480	2.38	-1389.812	56.923	MULTI SHOT	AAP	936374			
0	2008/11/17	8998	5182	-3540	90.370	177.210	1.22	-1418.431	58.250	MULTI SHOT	AAP	936375			
0	2008/11/17	9092	5182	-3540	89.750	177.800	0.91	-1447.055	59.497	MULTI SHOT	AAP	936376			
0	2008/11/17	9187	5183	-3541	89.070	178.350	0.92	-1475.993	60.470	MULTI SHOT	AAP	936377			
0	2008/11/17	9280	5185	-3543	88.270	178.070	0.91	-1504.318	61.355	MULTI SHOT	AAP	936378			
0	2008/11/17	9374	5187	-3545	89.880	178.000	1.71	-1532.947	62.337	MULTI SHOT	AAP	936379			
0	2008/11/17	9469	5187	-3545	90.000	178.310	0.35	-1561.888	63.269	MULTI SHOT	AAP	936380			
0	2008/11/17	9590	5187	-3545	89.380	178.490	0.53	-1598.754	64.299	MULTI SHOT	AAP	936381			
0	2008/11/17	9682	5189	-3547	88.830	178.530	0.60	-1626.783	65.028	MULTI SHOT	AAP	936382			

DEVIATION SURVEY REPORT BY FIELDWELL/LATERAL

FIELD AND WELL : ABUF-610 BOT.LATERAL NAME : 0 TIE-IN-DEPTH : 0 SERVICE COMPANY : SPERRY SUN				(WB : 0) TIE-IN-POINT : 936269				MAX: DRIFT ANGLE : 91.300 DEPTH(DF-FT) : 8810 VALID FLAG : Yes AAP FLAG : Yes				SURFACE LOCATION NORTHING : 2748019.55 EASTING : 193215.91 ZONE : 39 REF ELEV & CODE : 1642 DF			
LATERAL NAME	SURVEY DATE	MEAS DEPTH	TVD (DF)	TVD (SS)	DRIFT ANGLE(DEG)	TOOL DIRECT. (DEG)	DOGLEG (DEG/100 FT)	Y-OFFSET (METERS)	X-OFFSET (METERS)	TOOL TYPE	SURV. VERS.	DSPT NUM			
0	2008/11/17	9774	5190	-3548	89.510	178.780	0.79	-1654.813	65.686	MULTI SHOT	AAP	936383			
0	2008/11/17	9866	5191	-3549	89.810	178.100	0.81	-1682.844	66.450	MULTI SHOT	AAP	936384			
0	2008/11/17	9958	5191	-3549	89.940	178.280	0.24	-1710.871	67.336	MULTI SHOT	AAP	936385			
0	2008/11/17	10050	5192	-3550	89.070	178.050	0.98	-1738.897	68.233	MULTI SHOT	AAP	936386			
0	2008/11/17	10142	5193	-3551	88.830	178.960	1.02	-1766.924	68.965	MULTI SHOT	AAP	936387			
0	2008/11/17	10234	5194	-3552	89.810	179.410	1.17	-1794.961	69.364	MULTI SHOT	AAP	936388			
0	2008/11/17	10326	5195	-3553	89.750	179.300	0.14	-1823.000	69.679	MULTI SHOT	AAP	936389			
0	2008/11/17	10418	5195	-3553	89.630	179.650	0.40	-1851.040	69.936	MULTI SHOT	AAP	936390			
0	2008/11/17	10510	5196	-3554	89.320	179.480	0.38	-1879.080	70.149	MULTI SHOT	AAP	936391			
0	2008/11/17	10603	5197	-3555	89.940	178.770	1.01	-1907.422	70.582	MULTI SHOT	AAP	936392			
0	2008/11/17	10695	5197	-3555	89.810	177.490	1.40	-1935.448	71.497	MULTI SHOT	AAP	936393			
0	2008/11/17	10787	5198	-3556	88.890	177.170	1.06	-1963.457	72.803	MULTI SHOT	AAP	936394			
0	2008/11/17	10879	5199	-3557	89.260	177.610	0.62	-1991.466	74.080	MULTI SHOT	AAP	936395			
0	2008/11/17	10971	5200	-3558	90.250	177.770	1.09	-2019.484	75.210	MULTI SHOT	AAP	936396			
0	2008/11/17	11063	5201	-3559	88.640	177.670	1.75	-2047.501	76.326	MULTI SHOT	AAP	936397			
0	2008/11/17	11155	5202	-3560	90.310	178.250	1.92	-2075.523	77.324	MULTI SHOT	AAP	936398			
0	2008/11/17	11248	5202	-3560	89.320	178.130	1.07	-2103.854	78.219	MULTI SHOT	AAP	936399			
0	2008/11/17	11341	5203	-3561	89.750	177.770	0.60	-2132.182	79.233	MULTI SHOT	AAP	936400			
0	2008/11/17	11436	5203	-3561	89.880	177.880	0.18	-2161.117	80.332	MULTI SHOT	AAP	936401			
0	2008/11/17	11530	5205	-3563	87.650	177.150	2.50	-2189.732	81.574	MULTI SHOT	AAP	936402			
0	2008/11/17	11624	5208	-3566	89.260	177.200	1.71	-2218.337	82.985	MULTI SHOT	AAP	936403			
0	2008/11/17	11718	5208	-3566	89.940	177.110	0.73	-2246.952	84.407	MULTI SHOT	AAP	936404			
0	2008/11/17	11813	5208	-3566	90.370	177.240	0.47	-2275.873	85.834	MULTI SHOT	AAP	936405			
0	2008/11/17	11907	5208	-3566	89.440	177.070	1.01	-2304.488	87.257	MULTI SHOT	AAP	936406			
0	2008/11/17	12001	5209	-3567	89.320	177.160	0.16	-2333.102	88.699	MULTI SHOT	AAP	936407			
0	2008/11/17	12094	5210	-3568	89.510	177.550	0.47	-2361.416	90.007	MULTI SHOT	AAP	936408			
0	2008/11/17	12189	5211	-3569	89.570	177.180	0.39	-2390.341	91.338	MULTI SHOT	AAP	936409			
0	2008/11/17	12282	5211	-3569	89.750	177.170	0.19	-2418.652	92.735	MULTI SHOT	AAP	936410			
0	2008/11/17	12377	5212	-3570	89.940	177.450	0.36	-2447.576	94.094	MULTI SHOT	AAP	936411			
0	2008/11/17	12471	5212	-3570	89.510	177.380	0.46	-2476.198	95.386	MULTI SHOT	AAP	936412			
0	2008/11/17	12565	5213	-3571	88.830	176.890	0.89	-2504.810	96.818	MULTI SHOT	AAP	936413			
0	2008/11/17	12659	5216	-3574	88.210	177.080	0.69	-2533.412	98.324	MULTI SHOT	AAP	936414			
0	2008/11/17	12753	5219	-3577	88.150	177.390	0.34	-2562.015	99.706	MULTI SHOT	AAP	936415			
0	2008/11/17	12847	5222	-3580	88.520	177.430	0.40	-2590.625	101.000	MULTI SHOT	AAP	936416			
0	2008/11/17	12879	5222	-3580	88.270	177.480	0.80	-2600.365	101.433	MULTI SHOT	AAP	936417			
0	2008/11/17	12940	5224	-3582	88.000	177.500	0.44	-2618.930	102.247	MULTI SHOT	AAP	936418			

Well Location Map



Legend		 WellSites Division GIS	ABJF - 610
 EXPRESSWAY  MAJOR PAVED ROAD  GRADED ROAD  SKID ROAD (WSD)  SKID ROAD (SSD)  MAJOR TRACK  RAILWAY TRACK  PIPELINES	 POWERLINE  WELL  GOSP  CAMPSITE  HOSPITAL  POLICE/FIRESTATION		N: 2748019 E: 193215
1:100,000 (ZONE-39)			

THIS WELL LOCATION MAP AND THE INFORMATION CONTAINED HEREIN ARE THE SOLE PROPERTY OF SAUDI ARAMCO. NO REPRODUCTION IN FULL OR IN PART SHALL BE OBTAINED FROM THIS DOCUMENT WITHOUT WRITTEN CONSENT OF ITS OWNER. CREATED BY USING ArcGIS V8.2 (E&P GIS) AT MAPPING AND COMPUTATION FACILITIES - WELLSITES DIVISION - ABQAD - TEL: 572-5093 ON 07/12/10 7:00:00 AM

Table 5.1: Basic Pre-Intervention Well Data

Basic Pre-Intervention Well Data	
Connections	All tree and wellhead connections – complete schematic of all components required
Casing Weights, Grades, Depths and Dimensions (nominal & drift)	Best provided in a wellbore schematic
Tubing Weights, Grades, Depths and Dimensions (nominal & drift)	Best provided in a wellbore schematic and a table in the program
Wellbore Schematic	Legible schematic with sufficient detail
Deviation Survey	Survey of final wellbore including MD, TVD, Azimuth, Dogleg Severity, Vertical Section, degree deviation, direction
Flow Control Equipment Dimensions and Depths	Best provided in a wellbore schematic Historical problem and/or restriction information
Downhole Equipment	Details on packers, sliding sleeves, nipples, etc.
Elastomers	Define location of ALL elastomers together with type and compatibility with all anticipated fluid exposure (e.g. methanol, glycol, amine, etc.)
Wellbore Environment	Anticipated existing fluids - Possible reservoir fluid entry - Introduction of any fluids during intervention
Corrosive Well Fluids	CO ₂ , water, H ₂ S, chlorides, acid, etc.
Erosive/Corrosive Well Solids	Scale, sand, salt, hydrates
Well Pressure & Temperature	Surface and bottom hole, profiles/calculations
Well Logs	Open hole and/or cased hole as required for job
Intervention/Service Equipment Sizes	Dimensions, weight
Intervention/Service Equipment Capacity	Operating limits – pressure, temperature, running, pulling, overpull Lifting requirements like cranes, masts, etc.



Note: Master Copy of the Saudi Aramco Wireline Manual is maintained in electronic form.

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Table 5.2: Tools Check List

Tools Check List Electric/Braided Wireline Operations			
Function tested tools at surface prior to mobilization or RIH	Y	N	NA
Tools calibrated at surface prior to RIH	Y	N	NA
Tool string suitable for objectives – fit for purpose? (Sour Specifications/Conditions)	Y	N	NA
Any memory tools properly set – batteries OK incl unplanned time	Y	N	NA
Tightened tools in vertical position – Install pins to prevent backoff	Y	N	NA
Properly sized and configured tools for well conditions (centralizers, weight, flex joints, swivels, etc.)	Y	N	NA
Dedicated pressure testing equipment (HP, low volume)	Y	N	NA
Basic fishing tools available and present on wireline truck at the wellsite	Y	N	NA
Braided line only – mechanical & hydraulic jars in good condition	Y	N	NA
Complete diagram of tool string (diameters, lengths, etc) PRIOR to running in hole	Y	N	NA



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Electric and Braided Wireline Equipment and Operations
Section 8: Operational Procedures

Figure 8.2: Safety Guidelines

SAFETY GUIDELINES for WIRELINE/PERFORATING OPERATIONS				
Field Name		Well No.	Service Company	
Date		Service Company Rep		
		Saudi Aramco Engineer		
		Site Foreman/Supervisor		
ALL following safety steps must be completed prior to starting the operation			YES	NO
Any waivers must obtain approval from your Supervisor			NA	
1	Service Company temporary reflective direction signs on the highway Required for additional Emergency Assistance			
2	Access road to the well site is clear and unobstructed			
3	Flare lines to pit are safely secured			
4	Wireline and Control Equipment are positioned up wind from the wellhead Confirm with flag or wind sock If wind shifts into an unsafe condition – Shut down operations – Reposition equip			
5	Confirm crane and crane operator certification			
6	Establish safe area for unnecessary personnel or emergencies Position and mark at least 150 ft from work area Non-essential vehicles parked on pad edge, well away from flare lines/pit			
7	Install "NO ADMITTANCE" and "NO SMOKING" signs & tape around restricted area			
8	Night work must have a minimum of 5-7 light plants for gas operations and 4 for oil operations. Rig up NOT allowed at night if required light not available			
9	Truck communication fully operational Do NOT depend on another "group's" radio (because no control of malfunction) Critical jobs must have a mobile/satellite phone along with listing of critical phone numbers			
10	Personal Safety Equipment available			
11	Fire extinguisher(s) available and strategically positioned			
12	SCBA available and in the ready to use position: GAS 10-15 and OIL 4-6			
13	At least 5 H ₂ S sensors available for gas operations and 2 H ₂ S sensors available for oil operations			
14	Stretcher available and in the ready to use position			
15	Wheel chocks used			
16	Pre-Job safety meeting Objective Smoking Regulations Vehicle Parking Area	Responsibilities Safety Equipment Emergency Procedures	Potential Safety Hazards Job Procedures Pressure Test Operations	
17	Verified wireline and crane personnel training certificates			
18	Only Saudi Aramco personnel to operate tree valves (Count & Record valve turns)			
19	End of job clean up – Ensure well is secure and ALL fittings/equipment is removed			
Note: Do NOT begin job if above not completed				
WIRELINE				
20	Wireline retrievable SSSV has been removed (if applicable)			
21	Secure surface safety valve and tubing retrievable SSSV in locked open position (if applicable)			
22	Inside diameter and length of wireline lubricator and BOP will accommodate ALL tools			
23	Wireline BOP rams function & pressure tested with appropriate rod			
24	Lubricator system including grease injection function & pressure tested			
PERFORATION (in addition to wireline above)				
25	Cathodic protection is disconnected from the wellhead			
26	Measure for stray currents between the wellhead and wireline truck & ensure acceptable			
27	Is a "safe" firing system required and being used? → If so – next 2 steps not required (28 & 29) (Baker – GUARDIAN ; Halliburton – R.E.D. ; Schlumberger – SECURE)			
28	Ensure ALL communications radios & mobile phones are OFF			
29	Logging Engineer possesses the safety key prior to arming the gun until 250 ft in well Pulling out of well – from when gun reaches 250 ft until gun is disarmed Do NOT arm guns in a sandstorm			
30	Gun to CCL connections has NEW screws			
31	Follow ALL Service Company safety precautions			



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Electric and Braided Wireline Equipment and Operations
Section 5: Job Planning

Table 5.3: Pressure Control Equipment Check List

Pressure Control Equipment Check List Electric/Braided Wireline Operations			
Well Name		Job Date:	
Wireline Company		Wireline Engineer:	
		Wireline Operator:	
Saudi Aramco Supervisor			
Overall			
Compliance with Well Control Requirements by Well Type (Section 6 – Figure 6.1)	Y	N	NA
Rating compliance with maximum expected pressure and SOUR Service	Y	N	NA
Certification documents & history/maintenance records reviewed	Y	N	NA
Pressure tested in excess of maximum expected pressure	Y	N	NA
Well head connections ready (flanged for ALL oil & gas)	Y	N	NA
BOP			
Correct type and number of rams (only bottom wire ram pair inverted) As per Section 6 – Figure 6.1	Y	N	NA
Function & pressure test with calibrated rod (same rod for stuffing box)	Y	N	NA
Reviewed last maintenance – OK?	Y	N	NA
Connections, seal areas and O-rings in good condition	Y	N	NA
Accumulator capacity for one complete cycle for ALL rams (Close-Open-Close) Back up system ready	Y	N	NA
Lubricator			
Internal diameter and length appropriate for operation (entire tool string)	Y	N	NA
Thorough inspection of total riser length (damage, corrosion, etc)	Y	N	NA
Tool trap, tool catcher, stuffing box & line wiper function tested – OK?	Y	N	NA
Quick Test Sub (QTS) or full opening Safety Valve for perforating guns	Y	N	NA
Connections, seal areas and O-rings in good condition	Y	N	NA
Grease Sealing System			
Flow tube diameters & lengths (also check condition)	Y	N	NA
Grease flow rate adequate >2 liters/min @ pressure (prefer 4 lpm)	Y	N	NA
Adequate number of grease flow tubes (per Section 6.2.3)	Y	N	NA
Dual grease injection ports (each at different elevation)	Y	N	NA
Pressure monitoring at grease head injection point	Y	N	NA
Hydraulic control module function tested	Y	N	NA
Alternative grease pumping options (contingency)	Y	N	NA
Grease injection line check valve inspected and function tested	Y	N	NA
Grease selection suitable and in sufficient quantities for operation	Y	N	NA
Cable			
New cable – Torque released? – Speed limit on first unreeling	Y	N	NA
Cable suitable to handle loads – Condition good?	Y	N	NA
Cable is suitable for ENTIRE job, including any potential problems	Y	N	NA
Length on drum adequate	Y	N	NA
Sheave sizes appropriate (diameter 400 times largest strand size)	Y	N	NA
Inspected for flattening, external damage, stranding, torqued properly, etc	Y	N	NA
Clearly identify tension limits – Confirm weak point with Field Engineer	Y	N	NA
Miscellaneous			
Hydrate mitigation/removal – Glycol/Methanol pump available	Y	N	NA
Equalization capability – across swab valve, BOP rams, safety valve	Y	N	NA
Bleed off hoses secured & protected	Y	N	NA
Recording devices operational and good condition	Y	N	NA
Crane/mast capabilities and limitations	Y	N	NA
Spare valves, fittings, pressure gauges included	Y	N	NA



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Electric and Braided Wireline Equipment and Operations
Section 6: Equipment Standards, Specifications, Maintenance

Figure 6.1: Minimum Well Control Requirements by Well Type(see Notes below)
 SHALL be followed by ALL Service Providers.
 For the purpose of this manual, Saudi Aramco Wireline is considered a Service Provider.

	Gas Wells			Oil Wells		Water Injection			Water Source	
	LP	HP	Very HP	LP	HP	LP	HP	NP	Flow	No Flow
Valid Equip Cert Req'd	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
H ₂ S/CO ₂ Sour Service Req'd	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Max SIWHP (psi)	<5000	>5000	>10000	<1500	>1500	<1000	>1000	≤0	All	NA
WPR Rated Equipment (psi)	10,000	10,000	15,000+	5000	5000	3000	3000	3000	3000	3000
Equip Test Pressure-TP (psi)	7500	7500	Note(3)	Note(3)	Note(3)	Note(3)	Note(3)	Note(3)	Note(3)	Note(3)
Onsite Pressure Test Req'd	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pump #1→Hi Vol / Lo Press										
Pump #2→Lo Vol / Hi Press	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
(Service Company provided)										
Below Bubble Point	NA	NA	NA	No	Yes	NA	NA	NA	NA	NA
Knowledge of Gas Cusping/Coning	NA	NA	NA	No	Yes	NA	NA	NA	NA	NA
Near Populated Area (per SAES-B-62)	NA	NA	NA	No	Yes	NA	NA	NA	NA	NA
Total Barriers Required	3	3	3	2	3	2	2	1	1	0
Mechanical Barriers Req'd	2	2	2	1	2	1	1	1	1	0
WW Barrier Reduction (After bleed off → 24 hr SI <1000psi)	NA	NA	NA	NA	NA	NA	1	NA	NA	NA
Hydrostatic Barrier Margin (psi)	200	200	200	200	200	200	200	200	200	200
Observation Time for Hydrostatic Barrier (hr)	1	1	1	1	1	1	1	0	½	0
Grease Injection System	Hydraulic	Hydraulic	Hydraulic	Air	>5000 Hyd. (10)	Air	Air	Air	Air	Air
Line Wiper	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stuffing Box (Hydraulic)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grease Injection Tubes					>5000 (11)					
Above Upper Injection Point	3	5	5+	3	3	3	3	3	3	2
Injection Points Req'd	2	2	2	1	2	1	1	1	1	1
Below Lower Injection Point	2	2	2	1	2	1	1	1	1	1
Ball Check Valve	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Chemical Injection Sub with 15K Side Port Piloted Check Valve Assembly (on CIS)	Yes	Yes	Yes	No	No	No	No	No	No	No
Tool Catcher	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lubricator Hammer Unions	No	No	No	No	No	No	No	No	No	No
Lightweight Lubricator with O-Ring Seal & Premium Thread Connection	Yes	<7500 psi (12)	No	Yes	<7500 psi (12)	Yes	Yes	Yes	Yes	Yes
Lubricator with Integral Connection	Prefer	>7500 psi (13)	Yes	No	>7500 psi (13)	No	No	No	No	No
Bleed Off Sub or 2 Lubricator Bleed Offs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lubricator ID/Length for COMPLETE tool string	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tool Trap	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quick Test Sub (Perf ONLY)	Yes	Yes	Yes	No	Prefer	No	No	No	No	No
Single Ram BOP	No	No	No	No	No	No	No	No	No	No
Dual Ram BOP	No	No	No	Yes	<5000 (14)	Yes	Yes	Yes	Yes	Yes
Triple Ram BOP	Yes	Yes	No	No	>5000 (15)	No	No	No	No	No
Quad Ram BOP	No	No	Yes	No	No	No	No	No	No	No
Bottom Ram [Shear/seal] (17)	Yes	Yes	Yes	No	>5000 (16)	No	No	No	No	No
BOP ID for ALL tool string	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Multi-Line Ram Seal Inserts	No	No	No	Optional	Optional	Optional	Optional	Optional	Optional	Optional
BOP (Hydraulic/Manual)	Hyd	Hyd	Hyd	Hyd	Hyd	Man	Man	Man	Man	Man
Accumulator	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Hose BOP Equalization	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Block BOP Equalization	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Pump In Sub	NO → Use TKV on Tree			Yes	Yes	Yes	Yes	Yes	Prefer	Prefer



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NOTES: Well Control Requirements by Well Type (Figure 6.1)

1. Test Pressure Rating (TP_R) – Equipment

The maximum test pressure specified by the manufacturer, which SHALL not be exceeded at any time. The initial application of this test pressure will be applied to the equipment by the manufacturer prior to shipment. Subsequent tests are performed in suitable test bays (off location). The primary test is always a “shell” test of the body and not across any internal components of the equipment.

$TP_R = (2 \times WP_R)$ for Pressure control equipment rated 5000 psi or less

$TP_R = (1.5 \times WP_R)$ for Pressure control equipment rated for greater than 5000 psi

2. Working Pressure Rating (WP_R) – Equipment

The maximum working or operating pressure specified by the manufacturer which SHALL NOT be exceeded during any field operations.

3. Equipment Test Pressure (TP)

This pressure test is performed onsite prior to running in the hole operations.

Equal to the **GREATER of:**

$TP = (1.25 \times \text{Max WHP})$; 1.25 times the MAXIMUM Potential Wellhead Pressure and the resultant pressure SHALL NOT exceed the equipment Working Pressure Rating (WP_R). This allows an elevated pressure margin to facilitate bullheading of liquids into the wellbore during kill operations.

OR

$TP = (0.25 \times WP_R)$; 25% of the Equipment Working Pressure Rating (WP_R) which is only applicable in low pressure wells.

4. Equipment Working Pressure (WP)

Maximum allowable WP SHALL be EQUAL to Equipment Test Pressure (TP)

5. Normal Spot and Preventative Maintenance – Equipment

During continuous well-to-well field operations, the Service Provider SHALL take their truck and equipment back to the workshop/yard for normal spot and preventative maintenance after EVERY 5 wells. Severe service operations (e.g. severe corrosion) may dictate a more frequent maintenance requirement.

6. Manufacturer or Service Provider Regular Preventative Maintenance

The Service Provider SHALL perform spot and preventative maintenance at regularly scheduled intervals recommended by the manufacturer or by the Service Provider's in-house procedures. Records must be available for onsite inspection.

7. Compliance

ALL Service Providers, including Saudi Aramco Wireline, SHALL follow the above “Well Control Requirements by Well Type”. For the purpose of this manual, Saudi Aramco Wireline is considered a Service Provider.



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8. Definitions

- a. **Yes** – means item is required
- b. **No** – means item is NOT required
- c. **Prefer** – means item is preferred BUT NOT required
- d. **Optional** – means that the item can be used as an alternative

9. Equipment that Exceeds Minimum Requirements of Figure 6.1

Equipment that exceeds the minimum requirements for a given well type MAY be used.

Examples:

10K BOPE is acceptable for use on low pressure oil and water wells.

Triple BOP's are acceptable for use on wells that require a double BOP.

10. Grease Injection System for HP Oil Wells

If the SIWHP is greater than or equal to 5000 psig then a hydraulic grease injection system is required, otherwise an air system is acceptable.

11. Grease Injection Tubes for HP Oil Wells

If the SIWHP is greater than or equal to 5000 psig then use the number of tubes listed in this box, otherwise use the number of tubes listed for LP Oil Wells.

12. Lightweight Lubricator for HP Gas Wells or HP Oil Wells

If the SIWHP is less than 7500 psig then the lightweight lubricator is acceptable. If the SIWHP is greater than or equal to 7500 psig then a lubricator with integral connections is required.

13. Lubricator with Integral Connections for HP Gas Wells or HP Oil Wells

If the SIWHP is greater than or equal to 7500 psig then a lubricator with integral connections is required. If the SIWHP is less than 7500 psig then a lightweight lubricator is acceptable.

14. Dual Ram BOP for HP Oil Wells

A dual ram BOP is acceptable for oil wells with a SIWHP less than 5000 psig.

15. Triple Ram BOP for HP Oil Wells

A triple ram BOP is required if the SIWHP is greater than or equal to 5000 psi.

16. Bottom Ram (Shear/Seal) for HP Oil Wells

An additional ram is required if the SIWHP is greater than or equal to 5000 psig. This ram is in addition to the normal BOP requirement.

17. Bottom Ram (Shear/Seal)

This ram is in addition to the normal BOP requirement

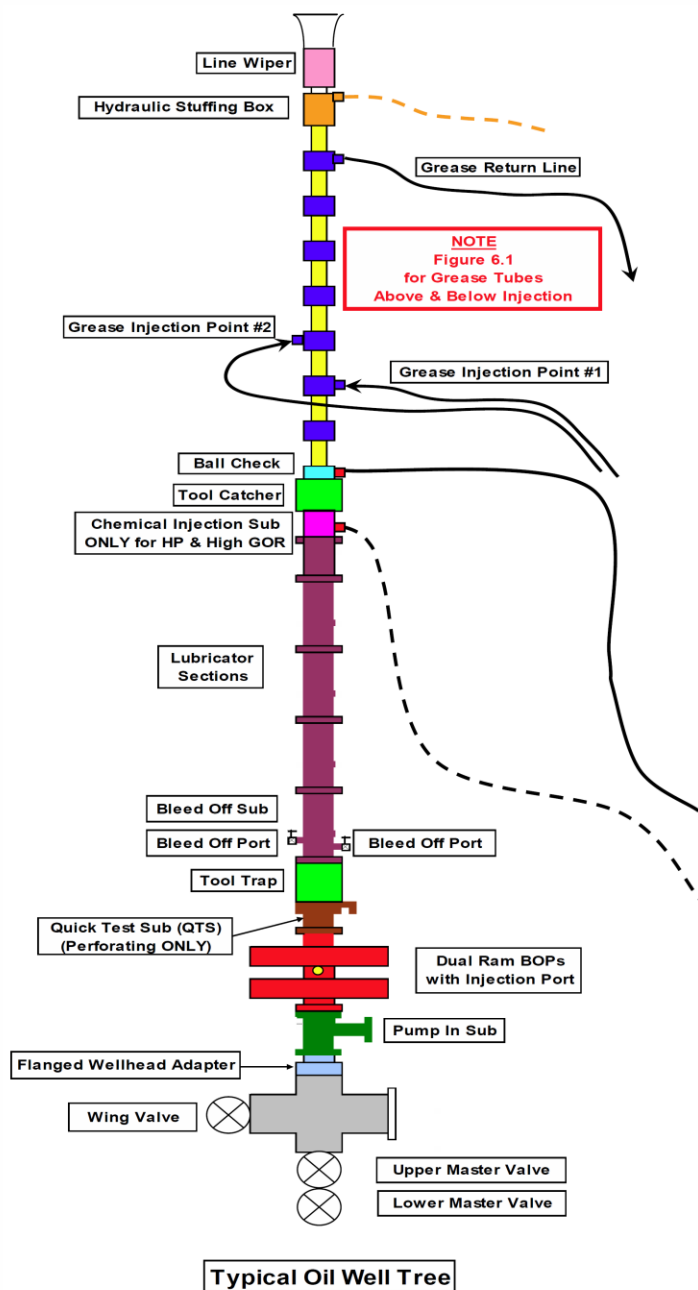


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Figure 6.2: Typical Electric/Braided Line Rig Up for Oil/Water Wells.



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